OWAIS AHMAD

DATA ENGINEER

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with professional experience designing and delivering ETL/ELT pipelines, Lakehouse architectures, API integrations, and analytics-ready datasets. Proficient in Python, SQL, PySpark, Apache Spark, Databricks, Delta Lake, Azure Databricks, FastAPI, PostgreSQL, and Docker. Experienced in Bronze-Silver-Gold Medallion Architecture, incremental data processing, data quality checks, REST API development, and BI reporting using Power BI and Databricks SQL. Passionate about building scalable, reliable data infrastructure and integrations that drive business intelligence and decision-making.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Engineering: ETL/ELT Pipelines, Data Ingestion, Data Cleaning, Data Transformation, Data Validation, Incremental Processing, Data Quality Checks, Batch Processing

Big Data & Lakehouse: PySpark, Apache Spark, Databricks, Azure Databricks, Spark SQL, Delta Lake, Medallion Architecture (Bronze/Silver/Gold), Auto Loader, Delta MERGE, Unity Catalog

Cloud Platforms: Azure Databricks, Azure Data Lake (exposure)

Databases & Warehousing: PostgreSQL, MySQL, SQLite, Data Warehousing Concepts, Databricks SQL, Spark SQL

BI & Visualization: Power BI, Databricks SQL Dashboards, Databricks AI/BI

Orchestration & DevOps: Databricks Workflows, Docker, Scheduled Background Jobs, Retry Handling

API & Integration: FastAPI, REST APIs, OAuth 2.0, LinkedIn API, Odoo API, JDBC

Development Tools: GitHub, Jupyter Notebook, Visual Studio Code, Streamlit, Microsoft Excel

PROFESSIONAL EXPERIENCE

Data Engineer | **YBIX** | *Lahore, Pakistan | May 2026 – Present*

- Build and maintain production-grade ETL/ELT pipelines using Python, PySpark, Databricks, and Spark SQL to process structured data for analytics and reporting workflows.
- Design and implement Bronze, Silver, and Gold Lakehouse layers using Delta Lake and Medallion Architecture, enabling organized and scalable data processing.
- Perform data profiling, cleansing, transformation, validation, and quality checks to ensure consistency and reliability of downstream datasets.
- Implement Delta Lake MERGE operations for incremental data loading and upserts to curated Gold-layer datasets.
- Create analytical datasets, KPI dashboards, and business reports to support data-driven decision-making across teams.
- Develop reusable, modular, and maintainable data engineering workflows; document pipeline architecture and data flows.
- Collaborate cross-functionally with data analysts and business stakeholders to translate requirements into scalable data solutions.

DATA ENGINEERING PROJECTS

End-to-End Data Engineering & AI Pipeline

Technologies: *Databricks, PySpark, SQL, Delta Lake, Auto Loader, JDBC, Unity Catalog, Databricks Workflows, Databricks SQL*

- Built a fully automated end-to-end data pipeline using Databricks, PySpark, Delta Lake, and Medallion Architecture (Bronze/Silver/Gold), handling ingestion through analytics-ready output.
- Implemented automated ingestion using Auto Loader (file-based) and JDBC (relational sources), enabling scalable and fault-tolerant data intake.
- Developed Bronze, Silver, and Gold Delta tables with data cleansing, validation, transformation, deduplication, and business-rule enrichment.
- Designed metadata-driven and scheduled Databricks Workflows for orchestration, pipeline monitoring, error handling, and alerting.
- Applied Unity Catalog and secret management for governed, secure data access; developed Databricks SQL dashboards and integrated AI/BI for natural-language analytics.

FMCG Data Engineering & Analytics Pipeline

Technologies: *Databricks, PySpark, Spark SQL, Delta Lake, Medallion Architecture*

- Designed an end-to-end FMCG data pipeline with Bronze, Silver, and Gold layers — covering cleansing, validation, transformation, deduplication, and business-rule processing.
- Built curated Gold-layer datasets for sales, products, customers, and business performance analysis.
- Implemented incremental processing using Delta Lake MERGE operations and applied Spark optimization techniques for scalable, maintainable analytics workloads.
- Created SQL-based KPIs and interactive Databricks SQL dashboards to monitor sales performance and operational metrics.

Online Retail Lakehouse & Analytics Pipeline

Technologies: *Azure Databricks, PySpark, SQL, Spark SQL, Delta Lake, Power BI*

- Processed ~541,909 retail transaction records through Bronze, Silver, and Gold layers on Azure Databricks; reduced to 527,390 valid records after cleansing, deduplication, cancellation filtering, and missing-value treatment.
- Implemented incremental processing using Delta Lake MERGE operations to maintain curated, up-to-date datasets.
- Developed Gold-layer analytical tables for revenue, product, customer, and sales KPIs; connected curated data to Power BI for interactive reporting.

Automated ETL Pipeline

Technologies: *Python, SQL, PySpark, Databricks*

- Designed and developed automated ETL workflows for extraction, transformation, validation, and loading with built-in data quality checks and monitoring.
- Used PySpark and Databricks for distributed batch data processing, improving pipeline reliability and repeatability.

Anviz Attendance Data Integration Pipeline

Technologies: *Python, FastAPI, PostgreSQL, Docker, Odoo API, ETL*

- Built an automated ETL pipeline that collects employee punch records from Anviz biometric devices via API, transforms and validates the data, and stores it in PostgreSQL — eliminating manual attendance exports.
- Developed REST API integration with Odoo ERP to synchronize employee attendance records, including employee ID mapping, duplicate prevention, and synchronization status tracking.
- Implemented scheduled background synchronization, retry handling for failed requests, and containerized the full solution using Docker for consistent, portable deployment.

LinkedIn Connection Tracker

Technologies: *Python, FastAPI, LinkedIn API, OAuth 2.0, PostgreSQL, REST API*

- Built a LinkedIn connection tracking application using Python and FastAPI, implementing OAuth 2.0 authentication to securely connect with the LinkedIn API.
- Developed connection lifecycle management tracking Pending, Accepted, Rejected, and Withdrawn states, with database persistence for full connection history.
- Added a dashboard with summary statistics and automated testing to ensure API reliability and data accuracy.

AI Personal Assistant

Technologies: *Python, FastAPI, Streamlit*

- Developed an AI-powered personal assistant application with FastAPI REST endpoints and a Streamlit interface for intelligent workflow automation.

EDUCATION

Bachelor of Science in Computer Science — Specialization in Data Science

University of Engineering & Technology (UET) Mardan | 2026 | CGPA: 3.00 / 4.00

CERTIFICATIONS & TRAINING

Databricks

- Databricks Data Engineering Training | Databricks Academy
- Databricks Training for Snowflake | Databricks Academy

IBM / Coursera

- Introduction to Data Engineering — Feb 2025 | IBM | Credential ID: UKQRSDQUEUUA
- BI Dashboards with IBM Cognos Analytics and Google Looker — Feb 2025 | IBM | Credential ID: 5TQDTAHDYCP4
- Data Analytics and Business Intelligence — Feb 2024 | IBM | Credential ID: PBQRG7QMK
- Introduction to Data Mining — Dec 2024 | Certificate Code: 7647754
- Get Started with Data Engineering | IBM

Artificial Intelligence & Machine Learning

- Artificial Intelligence & Machine Learning | KPITB (Khyber Pakhtunkhwa Information Technology Board)